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Further paleomagnetic results from Pleistocene sediments in northern Canada supporting the existence of stationary secular variation sources

M.E. Evans, K.P. Gillen and M.A.H. Hedlin *

Institute of Earth and Planetary Physics, University of Alberta, Edmonton, Alb. T6G 2J1 (Canada)

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Our recently published results from Pleistocene sediments in northwestern Canada provided no compelling evidence for the so-called Blake Event but did yield an interesting secular variation pattern interpreted as the signature of one or more stationary sources near the core–mantle boundary. New data, from a site about 1 km away, confirm this interpretation and thus support the division of the non-dipole field into drifting and standing parts in the geological as well as the historic past.

Measurements of the initial susceptibility of all 694 samples in this study and the previous one greatly facilitated stratigraphic correlation between the four 6 m sections involved.

1. Introduction

A great deal of paleomagnetic research is currently being carried out on recent sedimentary sequences around the globe. This activity reflects the need for temporally ordered data sets covering appropriate time intervals to define better the long-term behaviour of the geomagnetic field. As part of this effort, Hedlin and Evans (1987, herein referred to as paper I) investigated two 6 m sections of Pleistocene sediments exposed in the banks of the Old Crow River in the extreme northwest corner of Canada (139°48′W, 67°51′N). Here we extend this work in two ways: first, by describing additional results from two more 6 m sections; and, second, by determining the initial susceptibility of samples from all four sections and using this to aid stratigraphic correlation between them.

The results reported in paper I consist of natural remanent magnetization (NRM) and alternating field (AF) demagnetization measurements from a total of 338 samples collected in pairs at about 6 cm vertical intervals. Section A consists of 90 pairs, and section B (which is located 20 m upstream) consists of 79 pairs. Sections C and D, reported here, were collected similarly and consist of 91 and 87 pairs respectively. Section D is located 30 m downstream from section C, and these two are situated about 1 km from sections A and B. The age of these sediments is currently not well constrained; what information is available is discussed in paper I. All four sections are definitely bracketed by younger and older limits of 39 000 yr and 1.2 Ma respectively. The presence of the Old Crow Tephra in three of the sections indicates an age of ~ 100 000 yr.

* Present address: Scripps Institution of Oceanography, University of California, San Diego, La Jolla, California 920993, U.S.A.

Susceptibility was determined with a Bartington Instruments model M.S.1 calibrated by means of a series of chemical standards; all other field and laboratory procedures are described in paper I.

2. Results

Ten pilot samples were subjected to progressive AF demagnetization, and on the basis of the observed behaviour it was decided to clean all samples at both 10 and 20 mT peak fields. This is similar to the procedure adopted in paper I, and the combined results are summarized in Table I (rows 3–5). Between 0 and 10 mT there was a slight overall movement away from the present field direction at the sampling locality, but the 10 and 20 mT averages were indistinguishable. Further analysis was therefore carried out on the 10 mT data. Magnetograms of the mean declination and inclination at each horizon for sections C and D are given in Fig. 1. Sites for which the angular separation between the two individual sample vectors exceeds 10° have been rejected, as was done in paper I. This criterion is arbitrary, but the main features of the magnetograms are not affected by this choice—it is simply an objective means of eliminating poorer data. The lack of results in the lower part of section C is due to the very weak remanence carried by these samples which made it

TABLE I

Directional statistics for various data groups

		D	I	α_{95}	N
Present field		033.0	+82.0	–	–
Geocentric axial dipole		000.0	+78.5	–	–
ABCD	0 mT	356.5	+71.5	1.1	694
ABCD	10 mT	349.4	+68.9	1.3	622
ABCD	20 mT	348.9	+68.7	1.4	612
AB	All	356.3	+70.0	1.7	127
AB	Unperturbed	008.2	+72.3	1.7	75
CD	All	345.5	+66.1	2.5	105
CD	Unperturbed	344.2	+74.9	2.0	38
ABCD	All	351.0	+68.3	1.5	232
ABCD	Unperturbed	001.0	+73.5	1.4	113

D, I, α_{95} are declination, inclination and semi-angle of the cone of 95% confidence respectively (in degrees). N is the number of unit vectors (individual samples for rows 3–5; 10 mT AF cleaned horizon means for rows 6–11).

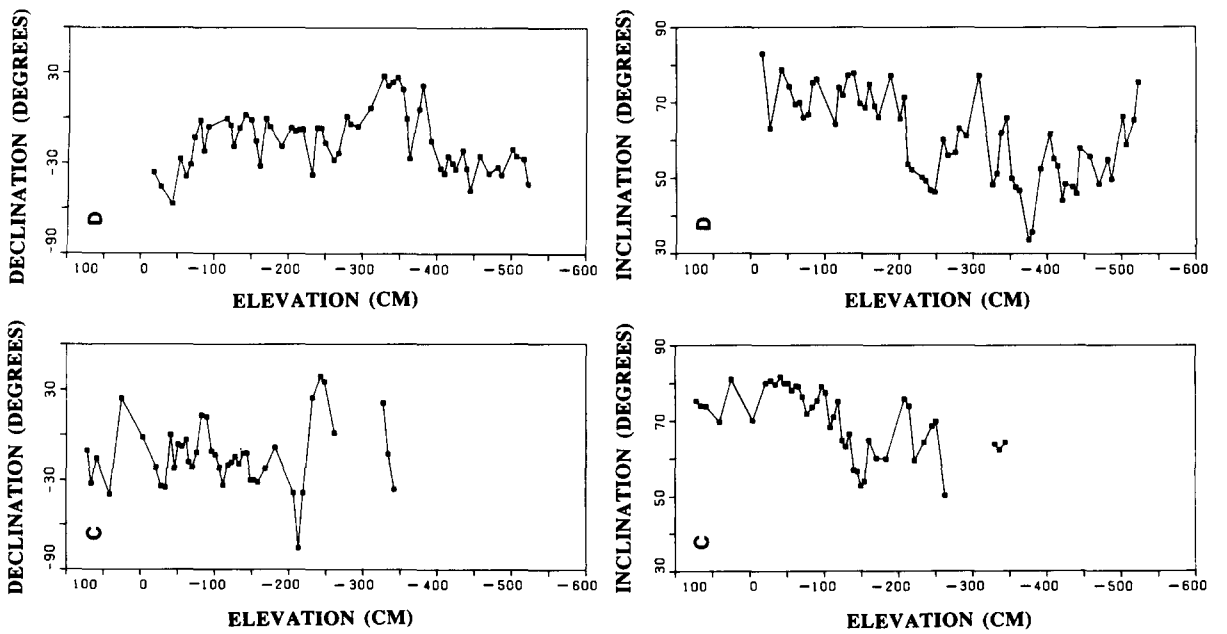


Fig. 1. Declination (upper) and inclination (lower) magnetograms for sections C and D. The gap near −300 cm in C is due to the presence of a 42 cm sand layer (see text and Fig. 2).

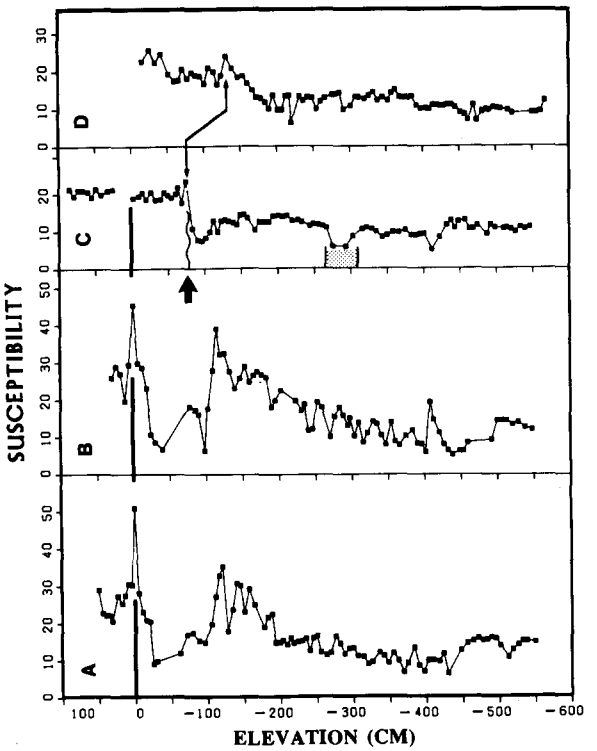


Fig. 2. Initial susceptibility (10^{-8} m^3kg^{-1}) profiles for sections A, B, C and D. The horizontal line at elevation zero in A, B and C represents the base of the Old Crow Tephra (2 cm thick in A and B, 15 cm thick in C); the Tephra is absent from section D, which consequently has an arbitrary zero. The marked maxima in A and B represent the Tephra itself but in C there are no corresponding data because the material was too friable to permit collection of coherent samples. The wavy line in C (accented by the bold arrow) represents an unconformity, and the stippled zone a sand layer.

impossible to measure them with sufficient internal precision. Thus the final data set consists of 105 horizons (44 in C, 61 in D).

Figure 2 illustrates the variation of initial susceptibility with depth in all four sections. Some strong correlations are immediately clear and it seems worthwhile to pursue the possibility of combining all four sets of magnetograms using the susceptibilities, the declination and inclination variations themselves, and the available geological control.

3. Discussion

Before analysing the secular variation patterns recorded by the four profiles, we point out that one of the original goals of this project was to seek evidence for the Blake Event, which has been proposed as a short interval of reversed polarity. No reversed samples were found in any of our sections, which suggests that one or more gaps exist in the sedimentation. Geologically, this is quite likely and, as the following discussion makes clear, the remanence and susceptibility data themselves support this suggestion.

3.1. Sections C and D

In the field there is a noticeable unconformity at a depth of −81 cm in section C and this is clearly reflected in the susceptibility pattern, which shows an abrupt drop by a factor of about two between depths of −76 and −83 cm. A similar

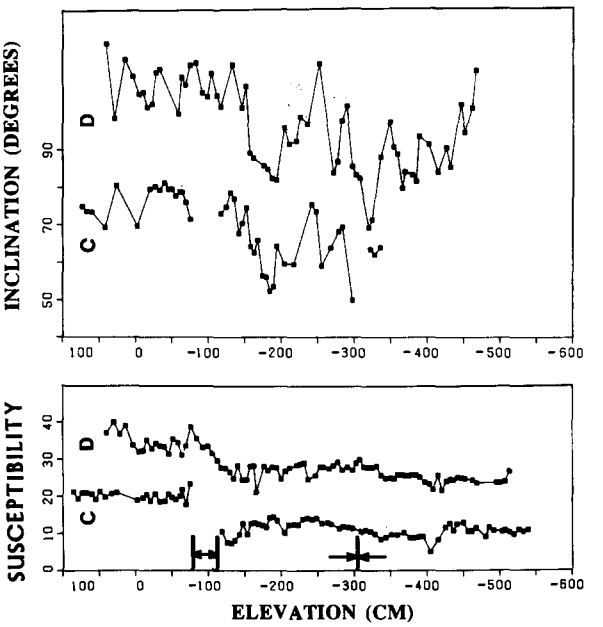


Fig. 3. Illustrating the correlation between sections C and D after the stratigraphic adjustments described in the text. The scales at the top refer to the data for section C. For clarity, susceptibilities for D have been moved arbitrarily to the right by 15 scale units and inclination values have been moved by 35°. The 35 cm gap inserted and the 42 cm sand layer removed are indicated by appropriately directed arrows.

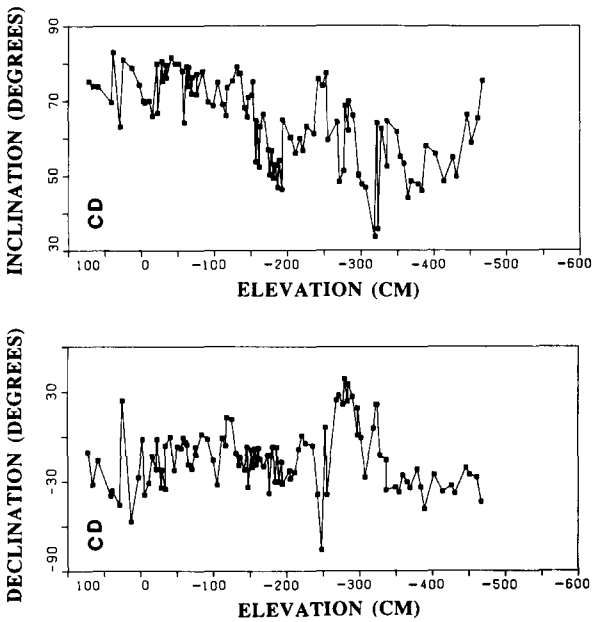


Fig. 4. Composite declination and inclination magnetograms for section CD.

drop occurs in section D, but in this case a more gradual, monotonic, decrease takes place over the 30 or 40 cm below −131 cm, suggesting that there is no unconformity present. To combine these two curves we follow this suggestion, first by shifting section D upwards by 55 cm to bring the horizon at −131 cm into line with a depth of −76 cm in section C, then by opening a gap of 35 cm representing the missing sediments in C. In addition, there is a 42 cm thick sand layer in C which has no counterpart in D. As sands are likely to have been deposited rapidly we have removed this layer entirely. The outcome of these three modifications is illustrated in Fig. 3, which also shows the corresponding inclination results. It is clear that there is now a strong correlation, and this is confirmed by the corresponding correlation coefficients. With no adjustments between C and D the correlation coefficient is 0.40, but this increases to 0.60 after the 55 cm shift and then to 0.86 after the 35 cm gap is inserted and the sand removed. A combined

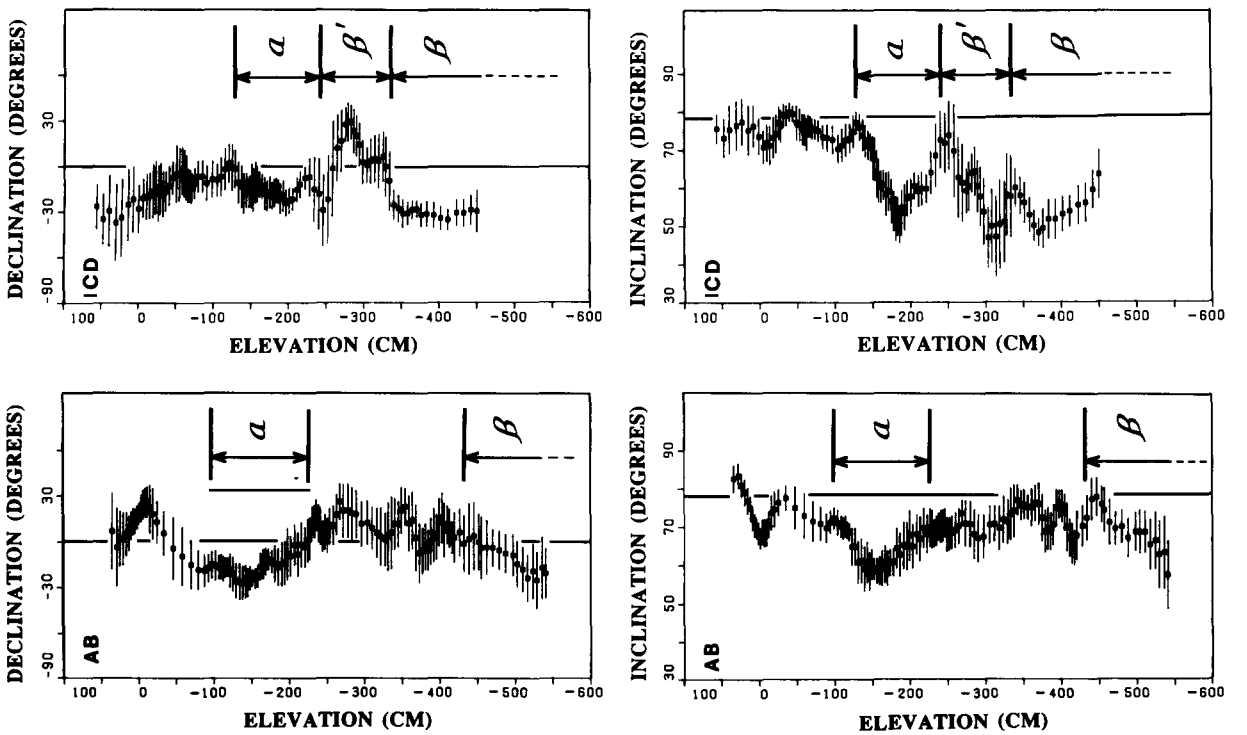


Fig. 5. Smoothed magnetograms for sections AB and CD. Horizontal bars represent α_{95} limits obtained from a moving 5-pair vector average. The geomagnetic perturbations α, β and β' discussed in the text are indicated. The vertical lines represent the declination (zero) and inclination (+78.5°) of a geocentric axial dipole.

CD section constructed by the above procedure is illustrated by the magnetograms of Fig. 4. The correspondence is very satisfactory, the vector correlation coefficient being 0.97.

3.2. Sections A and B

In paper I these two sections were combined on the basis of geological field observations by tracing recognizable beds between them. For sections C and D the visible correlation was much less clear and this was the incentive for attempting a correlation using susceptibility measurements as described above. When the geological correlation used in paper I is applied to the susceptibility data for sections A and B the correlation coefficient decreases from 0.86 to 0.79 as the geological unit boundaries in one section are adjusted to match those in the other. This is a minor decrease but it is worth enquiring into its cause. Inspection of Fig. 2 indicates that there is a broad susceptibility minimum lying between the Old Crow Tephra peak and the peak at about −120 cm elevation. This feature is present in both sections, but below the lower peak in A there is an abrupt drop whereas in B a more gradual decrease is seen. If a small gap is opened up to allow for the apparently missing sediments in A, the correlation coefficient increases from 0.86 to a maximum of 0.89 for a 10 cm gap. These details indicate that in favourable circumstances very fine-scaled matching of parallel sections may be possible, but the important point is that, in this case, correlation by geological field observations or by laboratory susceptibility measurements lead to insignificant differences in the final secular variation pattern. The combined, smoothed declination and inclination magnetograms shown in Fig. 5 are virtually identical with the corresponding data in Fig. 2 of paper I.

3.3. Sections AB and CD

We now turn to a comparison of these two composite sections. Referring to Fig. 5, we draw attention to the low inclination swing in AB between elevations −99 and −226 cm (labelled α), and the latter part of a similar swing (β) in the lowermost metre or so of the section. These were

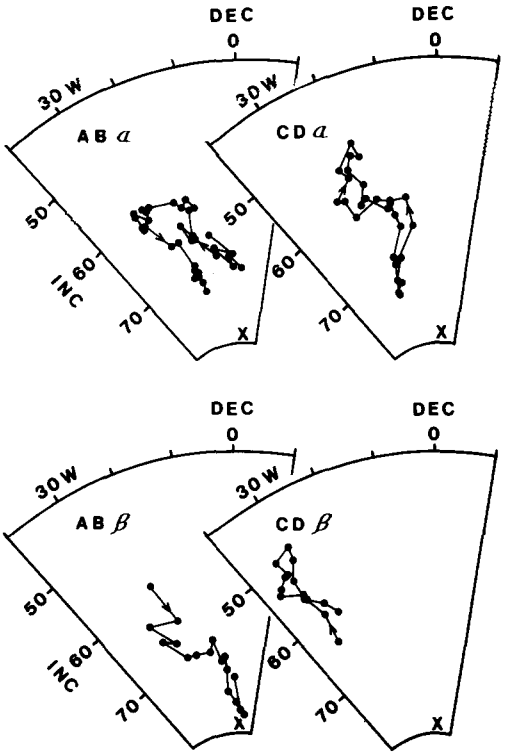


Fig. 6. Declination–inclination plots for the perturbations α and β recorded in sections AB and CD. Points plotted are 5-pair vector averages. Elevation ranges are as follows: AB(α) −99 to −226 cm; CD(α) −121 to −241 cm; AB(β) −434 to −542 cm; CD(β) −335 to −451 cm. Crosses indicate the field of a geocentric axial dipole.

discussed at length in paper I, where it was shown that they represent perturbations consisting of linear sequences of field vectors, as shown in Fig. 6. The CD data show remarkably similar behaviour (Figs. 5 and 6) and thus strongly support the reality of these patterns. Furthermore, the lowermost part of CD provides evidence of a more complete perturbation, consisting of both outward and return trajectories, whereas the original AB data provided only a partial record.

Although the two perturbations in both composite sections correlate very well it should be noted that the elevations do not directly correspond. This is best seen by inspecting the inclination data in Fig. 5. Starting at the top, we note that there is a small minimum near the Old Crow Tephra followed by ~ 1 m of sediments in which the inclination does not vary greatly. Below this

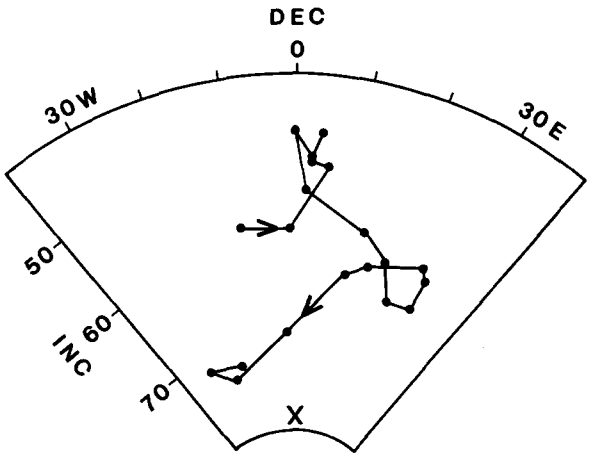


Fig. 7. Declination–inclination plot of the perturbation β' in section CD. Elevation range is −241 to −335 cm.

the upper perturbation (α) occurs. Then in section AB there are ~2 m of sediments with fairly steady inclination values before the lower perturbation (β) is seen (below about −430 cm). Section CD, however, lacks the steady inclination zone; rather, we see another marked low-inclination swing (β'). The polar zenithal equal area nets (Figs. 6 and 7) convincingly demonstrate the similarity of features α and β in both AB and CD, but the feature β' seen in CD has no counterpart in AB, nor is the long sequence of more or less constant inclinations present in AB found in CD. It appears that the interval between features α and β is not fully represented at both sites. The precise manner in which the beds pinch out laterally is not important here but it is of interest to note that β' differs from α and β in that it involves easterly declinations, whereas the others swing to the west of north. Furthermore, feature β' does not have the form of a linear perturbation arising from a stationary source. Instead, it traces out a clockwise loop (Fig. 7) reminiscent of westward-drifting sources (Runcorn, 1959; Malin and Bullard, 1981).

Table I (rows 6–11) lists various average vectors for different combinations of sections A, B, C, and D. In all cases, there are deviations from the present field and the field due to a geocentric axial dipole. Of course, part of this effect is a result of the perturbations themselves, but the deviations persist—in diminished form—even when the perturbed zones are removed. As discussed in paper I, this is most probably owing to the fact that the time interval spanned by our sampling is too short to average out the complete spectrum of secular variation, in particular, the dipole wobble. Proof of this suggestion must await better dating control of these sediments; for the moment we regard the mean directions of the 'unperturbed' zones as nothing more than a 'snapshot'.

4. Conclusions

One of the most vexing problems in this kind of work is the matching of parallel time sequences. Several authors (e.g., Thompson and Oldfield, 1986; Dearing et al., 1981) have found that initial susceptibility often serves as a useful parameter in this context, and this study provides further evidence that this is so.

The new remanence results reported here confirm the reality of the linear secular variation perturbations reported recently by Hedlin and Evans (1987) from two stratigraphic sections situated ~1 km away. There is now a growing body of evidence that these features are the signatures of stationary pulsating sources in the outermost core (Creer and Tucholka, 1983; Evans, 1984), and thus models of the secular variation involving both standing and drifting sources are realistic for the geological as well as the historic past (Yukutake and Tachinaka, 1969; Bloxham and Gubbins, 1985). Such sources need only achieve peak magnitudes of a few per cent of the central dipole to generate perturbations of 20–30° as found here and in other published examples.

Acknowledgments

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